

Project Direction and Verification Matrix

1. Executive Summary

This matrix consolidates the project's current research direction, benchmark position, verification status, and immediate work plan. It is intended to provide a concise entry point before the supporting documents are reviewed.

The project direction is to transform the current AI and knowledge-graph prototype into a verified research pipeline. Richmond et al. (2020) is used as the human benchmark. The AI-generated outputs will be evaluated at the levels of corpus coverage, entity mapping, relationship triples, CMOC structure, cross-study synthesis, and final programme theory alignment.

2. Direction Statement

The core direction is:

Benchmark against Richmond
-> verify corpus representation
-> verify entities and relationships
-> evaluate pipeline stages
-> compare final synthesis with Richmond
-> produce a defensible research evidence package

The intended research claim is:

This project evaluates whether an AI-assisted GraphRAG and multi-agent pipeline can reproduce key analytic operations of Richmond et al.'s human realist review, including corpus representation, entity/relationship modeling, CMOC extraction, synthesis, contradiction handling, and final programme theory comparison.

3. Verification Matrix

Project issue	Current position	Evidence base	Verification action
Final project goal	Produce and evaluate an AI-assisted realist synthesis pipeline against Richmond's human review.	Pipeline_Evaluation_Framework.md; current matrix.	Maintain focus on verified synthesis rather than graph visualization.
Richmond benchmark	Richmond provides a layered benchmark: 28 included studies, human realist workflow, five context-specific programme theory findings, final conclusion, and E01-E47/R01-R40 operationalization.	Richmond_Golden_Standard_Memo.md; data/gold_standard.json.	Use Richmond for corpus, theory, entity, relationship, and final synthesis comparison.
Available gold standard	Gold exists at the corpus, programme-	28-study registry; E01-E47; R01-R40; PTS1-	Create a human-adjudicated per-paper

Project issue	Current position	Evidence base	Verification action
	theory, entity, and relationship-structure levels. Full per-paper extraction gold is not yet available.	PTS5.	CMOC/entity/relation table after the initial verification package.
Richmond conclusion	Intervention effects depend on learner knowledge, confidence, self-efficacy, coping, feedback, and opportunities for clinical reasoning practice.	Richmond et al. (2020); <code>Richmond_Golden_Standard_Memo.md</code> .	Compare AI final synthesis against Richmond's five-context programme theory.
Entity definition	An entity is a Richmond-relevant realist synthesis concept: Context, Intervention, Mechanism Resource, Mechanism Response, or Outcome.	<code>Entity_Relationship_Definition_Modeling_Specification.md</code> ; E01-E47.	Normalize raw graph nodes to E01-E47 or classify them as rejected/new human-approved concepts.
Relationship definition	A relationship is a directed, typed, evidence-supported link between two normalized entities.	<code>Entity_Relationship_Definition_Modeling_Specification.md</code> ; R01-R40.	Export model edges as (<code>source_e_code</code> , <code>predicate</code> , <code>target_e_code</code>) triples.
Relation vocabulary	Final edge types are PROVIDES, ENABLES, TRIGGERS, LEADS_TO, and CONSTRAINS.	<code>data/gold_standard.json</code> ; entity/relationship specification.	Align code and output artifacts with the frozen relation vocabulary.
KG verification status	Preliminary outputs exist, but KG verification is incomplete.	Current gaps: 20-vs-28 GraphRAG issue, 56-vs-57 CMOC mismatch, no strict R01-R40 relationship F1.	Audit corpus coverage, reconcile CMOC count, and compute entity/relationship metrics.
AI-human comparison	Comparison will be performed at multiple levels: corpus coverage, entity mapping, relationship triples, CMOC structure, PTS coverage, contradiction handling, and final programme theory	<code>Pipeline_Evaluation_Framework.md</code> ; <code>Entity_Relationship_Verification_Protocol.md</code> .	Produce <code>KG_Verification_Report.md</code> , <code>Pipeline_Metrics_Report.md</code> , and <code>Richmond_Comparison_Memo.md</code> .

Project issue	Current position	Evidence base	Verification action
	alignment.		
Error refinement	Errors will be localized, categorized, corrected through prompt/schema/mapping updates, rerun, and re-scored.	HITL refinement loop in evaluation and verification documents.	Replace auto-approval behavior with human-readable review records for official evaluation.
Post-KG use	The KG supports CMOC extraction, cross-study synthesis, contradiction detection, and final programme theory comparison.	Current pipeline generates CMOCs, contradictions, demi-regularities, and programme theory text.	Verify the KG before using it as evidence for final comparison.
Immediate execution direction	Complete a 48-hour verification sprint.	Next_Step_Work_Plan.md.	Finish corpus audit, CMOC reconciliation, relationship triple export plan, and report structures.

4. Immediate Verification Sprint

The next work phase is a 48-hour verification sprint.

Priority	Deliverable	Function
P0	Professor_Response_Matrix.md	Concise project direction and verification matrix.
P0	outputs/corpus_coverage_report.md	Confirms representation of all 28 Richmond studies.
P0	CMOC reconciliation note	Resolves the official CMOC count.
P0	Relationship triple export plan	Defines conversion from current relationship outputs to R01-R40 scoring format.
P1	outputs/KG_Verification_Report.md draft	Prepares node and edge verification reporting.
P1	outputs/Richmond_Comparison_Memo.md draft	Prepares AI-vs-human synthesis comparison.

5. Supporting Documentation

The response matrix is the entry document. The supporting documents provide detail when needed.

Supporting document	Function
Professor_Project_Briefing.md	Concise project overview.
Richmond_Golden_Standard_Memo.md	Richmond benchmark definition.
Entity_Relationship_Definition_Modeling_Specification.md	Entity and relationship semantics.
Entity_Relationship_Verification_Protocol.md	Entity and relationship metrics.
Pipeline_Evaluation_Framework.md	Stage-by-stage evaluation design.
Current_Project_Status_Gap_Report.md	Current assets, gaps, and unsupported claims.
Next_Step_Work_Plan.md	Immediate verification work plan.
Selective_Research_Foundation_Map.md	Reference and tool selection policy.

6. Conclusion

The project has a defined direction and benchmark structure. The immediate requirement is to convert the current prototype and planning documents into verification evidence: corpus coverage, reconciled CMOC counts, relationship triples, KG metrics, and final Richmond comparison.